

Disturbance impacts on swamp and forested peatlands in southern Québec



Lead supervisor: Scott J. Davidson – Université du Québec à Montréal (UQAM)

Project overview:

This PhD project aims to quantify how anthropogenic disturbances - including agriculture, forestry, and road construction - alter carbon cycling and hydrological functioning in swamp and forested peatlands of southern Québec. By explicitly linking land-use history to greenhouse gas fluxes, belowground carbon storage, microbial dynamics, and water table variability, the project will assess how disturbance reshapes the coupled carbon–water processes that regulate ecosystem carbon balance.

Using measurements of CO₂ and CH₄ fluxes from vegetation, soils, and trees across forested wetlands with contrasting disturbance histories, combined with peat and soil coring, root sampling, microbial community analyses, and continuous hydrological monitoring, the student will evaluate how shifts in hydrological regimes and belowground processes mediate carbon flux responses. These plot-scale observations will be integrated to improve understanding of disturbance-driven changes in peatland carbon dynamics under ongoing land-use and environmental change.

Research objectives:

1. Quantify carbon flux responses to disturbance: measure CO₂ and CH₄ fluxes from vegetation, soils and trees across forested wetlands with contrasting land-use histories.
2. Evaluate peat and microbial responses: collect peat/soil cores, roots, and analyze microbial communities to assess how disturbance alters belowground carbon storage and processes.
3. Assess hydrological alterations: install hydrology loggers to monitor water table variability and link changes to carbon fluxes

Candidate Requirements:

- MSc degree in environmental science, ecology, geography, soil science, hydrology, biogeochemistry, or related field.
- Fieldwork and laboratory experience.
- Strong written and oral communication skills.

Training and opportunities:

The student will receive training in greenhouse gas flux measurements (CO₂ and CH₄) peat and soil coring techniques, microbial community analyses, hydrological monitoring, and integrated carbon–hydrology data analysis. The project offers opportunities for interdisciplinary collaboration across biogeochemistry, hydrology, and microbial ecology, engagement with land-use stakeholders, and participation in regional and international wetland research networks.

How to apply:

Please submit the following as a single PDF file to Scott J. Davidson davidson.scott_j@uqam.ca:

- Cover letter outlining research interests and motivation for the position
- Academic CV (including publications, field/lab experience, technical skills)
- Transcripts
- Names and contact information of two referees

Applications will be reviewed on a rolling basis until the position is filled.